Architecture Evaluation Report Template
Physiological Digital Twin CAPAR

Template ini digunakan untuk mengevaluasi kualitas Software Architecture Physiological Digital Twin berbasis CAPAR. Evaluasi dilakukan untuk memastikan arsitektur memenuhi kebutuhan stakeholder, quality attributes, traceability, dan kesiapan implementasi.

1. Informasi Evaluasi

Nama sistem:
Versi arsitektur:
Tanggal evaluasi:
Evaluator:
Metode evaluasi:
- ATAM (Architecture Tradeoff Analysis Method)
- Scenario-based evaluation
- Requirement traceability analysis

2. Tujuan Evaluasi

Tujuan:
1. Menilai apakah arsitektur mendukung kebutuhan CAPAR.
2. Mengidentifikasi risiko arsitektur.
3. Mengevaluasi trade-off antar quality attribute.
4. Memastikan hubungan requirement dengan komponen sistem.

3. Deskripsi Arsitektur yang Dievaluasi

Ringkasan arsitektur:

Layer 1:
Multimodal Acquisition Layer

Layer 2:
Governed Information Layer

Layer 3:
Analytical Evidence Layer

Layer 4:
Operationalization Layer

Komponen utama:
- Data Acquisition
- Feature Engine
- Baseline Engine
- CAPAR Engine
- Digital Twin Engine
- XAI Engine
- Dashboard

4. Stakeholder dan Concern

Template tabel:

Stakeholder:
Concern:
Expected quality:

Contoh:

Clinician:
Interpretability
Output harus memiliki evidence dan uncertainty.

Researcher:
Reproducibility
Model, data lineage, dan parameter harus tercatat.

Programmer:
Maintainability
Komponen modular dan terdokumentasi.

5. Quality Attribute Scenario Evaluation

Gunakan format:

Source:
Stimulus:
Environment:
Artifact:
Response:
Response Measure:

Contoh:

Source:
Wearable sensor

Stimulus:
Data hilang selama monitoring

Environment:
Longitudinal monitoring

Artifact:
Quality Gate Module

Response:
Sistem memberikan warning dan confidence reduction

Measure:
Missing data tolerance

6. Architecture Trade-off Analysis

Quality attribute yang dianalisis:

1. Accuracy
2. Explainability
3. Real-time performance
4. Personalization
5. Scalability
6. Security

Template:

Decision:
Trade-off:
Benefit:
Risk:
Mitigation:

7. Risk Identification

Template:

Risk ID:
Deskripsi risiko:
Komponen terkait:
Dampak:
Probabilitas:
Severity:
Mitigation:

Contoh risiko:

R-001:
Ketergantungan terhadap baseline personal yang belum matang.

Dampak:
False alert meningkat.

Mitigation:
Quality gating dan shrinkage baseline.

8. Requirement Traceability Evaluation

Mapping:

Requirement
|
Architecture Component
|
Implementation Module
|
Validation Method

Contoh:

FR-007 CAPAR State Engine

Component:
CAPAR FSM

Module:
state_transition.py

Validation:
Episode F1 score

9. Scenario-Based Evaluation

Scenario 1:
Continuous physiological monitoring

Scenario 2:
Recovery trajectory estimation

Scenario 3:
Relapse detection

Scenario 4:
Clinical-context fusion

Untuk setiap scenario:

Input:
Processing:
Expected output:
Architecture component involved:
Evaluation result:

10. CAPAR Specific Evaluation Criteria

Evaluasi khusus:

A. Physiological validity
- hubungan feature-latent variable

B. State validity
- makna fisiologis state

C. Personalization
- baseline adaptation

D. Context awareness
- activity-aware interpretation

E. Explainability

- evidence-based output

11. Evaluation Metrics

Metrics:

Performance:
- latency
- throughput

Model:
- F1
- sensitivity
- specificity
- calibration

State:
- FSM stability
- transition consistency

Recovery:
- TTR error
- trajectory similarity

XAI:
- explanation validity
- expert agreement

12. Architecture Strengths

Isi:

Kekuatan utama arsitektur:
1.
2.
3.

Bukti pendukung:

13. Architecture Weaknesses

Isi:

Kelemahan:
1.
2.
3.

Rencana perbaikan:

14. Final Evaluation Summary

Overall assessment:

[ ] Acceptable
[ ] Acceptable with modification
[ ] Major revision required

Kesimpulan evaluator:

Rekomendasi:

15. Approval

Evaluator:

Nama:
Tanggal:
Tanda tangan:

System Architect:

Nama:
Tanggal:
Tanda tangan: